

Verifying What Sensors Can and Cannot Close: A Statistical Observability-Boundary Study of a Bare-Metal, LLM-Free Autonomous Decision Executive

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Kwangwoon University — sensor-agnostic autonomy · decision core in portable C (no ROS / Linux / GPU) · scored by an independent physics referee

Document type: Technical Report (independent-V&V study). **Prepared by:** Kwangwoon University (not a NASA publication; NASA standards are cited as prior art). **Distribution:** unlimited.

Subject terms: autonomous decision executive; verification and validation; independent V&V; statistical observability; observational equivalence; cross-modal consistency; fault detection, isolation, and recovery; Kalman filtering; micro-Doppler liveness; search and rescue; bare-metal flight software.

Fidelity of results: all quantitative results are simulation verification at the measured-terrain level, reported with confidence intervals over hundreds of episodes; operational validation is enumerated as remaining work, not claimed.

Abstract — We treat a bare-metal, LLM-free autonomous search-and-rescue decision executive as an object of software Verification and Validation (V&V) rather than as a novel algorithm, and we give the word observability a precise statistical meaning: two hidden scenarios are observationally equivalent under an information set when they induce the same law over observation sequences, so that no discriminator can separate them; a boundary is the set of scenario pairs an onboard sensor suite can and cannot separate. This is a hypothesis-distinguishability notion, distinct from the control-theoretic observability matrix, which concerns state reconstruction from a known linear model. The executive composes established twentieth-century autonomy structures — subsumption-style arbitration, a blackboard world state, a three-layer executive with hierarchical planning, Kalman belief estimation with an innovation-consistency statistic, and fault detection, isolation, and recovery — into an operating-system-free C superloop, scored by an independent physics referee that holds hidden truth and evaluates any player through observations alone. Over hundreds of episodes with confidence intervals we show that temporal persistence, kinematic consistency, and fault management each reject a distinct false-signal class, yet leave one shared indistinguishable region: a stationary, kinematically consistent, persistent false target. Cross-modal physical consistency closes single-channel clutter without new hardware, at a measured cost to single-channel real targets; but a full-signature decoy remains observationally equivalent to a real target under the existing channels. We then recast liveness not as a single sensor but as the minimal additional observable set needed to make the target and decoy laws statistically distinguishable, and argue this set is environment-dependent: neither motion nor thermal contrast is by itself liveness, because wind-driven vegetation and sun-warmed terrain confound them. A cardiopulmonary micro-Doppler phase is shown, in the modeled regime, to be a strong candidate observable that separates the two laws, while identifying the physically realizable airborne observable that suffices under each environment remains an open sensing-design problem. Every result is labeled as simulation verification at a stated fidelity level, with validation kept separate. Negative results are kept.

Index Terms — autonomous systems, verification and validation, independent V&V, statistical observability, observational equivalence, sensor fusion, cross-modal consistency, fault detection isolation and recovery, Kalman filtering, micro-Doppler liveness, search and rescue, bare-metal software.

Nomenclature

$\mathbf{x}, \mathbf{P}$	state vector [position, velocity] and its error covariance	$\mathbf{l}$	information set (the multi-channel observation sequence)
$\mathbf{F}, \mathbf{Q}, \mathbf{H}, \mathbf{R}$	CV transition, process-noise, measurement, and measurement-noise matrices	H_T, H_D	hypotheses: real target vs. decoy
$\mathbf{K}, r, \mathbf{S}$	Kalman gain, innovation, innovation covariance	z	candidate additional physical observable (liveness variable)
d^2	normalized innovation squared, $d^2 = \mathbf{r}^T \mathbf{S}^{-1} \mathbf{r}$	D_{KL}	Kullback-Leibler divergence between two laws
γ, N	NIS gate threshold; temporal confirmation window (steps)	ϕ, λ	RF phase and carrier wavelength (micro-Doppler)
w_i, h_i, c_i	sensor vote weight, health, evidence (fusion)	A_r, f_r	respiration amplitude and rate; A_h, f_h for heartbeat
$U(a), \lambda, \mu, \alpha$	action utility; cost, energy, and NASA-rule weights	σ, n_{modal}	position uncertainty $\sqrt{\text{tr } \mathbf{P}}$; count of distinct agreeing channels

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1. Introduction

1.1 Motivation

A recurring assumption is that autonomous decision-making now requires a large language or foundation model. Historically this is false: NASA's Remote Agent flew an autonomous planning-scheduling-and-execution architecture with model-based fault protection on Deep Space One in 1999 [출처:조각]. Our premise is that software can structure observations and compose predefined decision primitives, and that the interesting question is not which algorithm is used but how far a given sensor suite and decision rule can be trusted — answered by verification, and by a precise account of what the sensors can and cannot distinguish.

1.2 Approach

We build a decision executive from the classical autonomy lineage, subject it to an independent physics referee, and ask the referee to draw an observability boundary in the statistical sense of Section 6.1: which pairs of hidden scenarios the onboard information can separate, and which pairs remain observationally equivalent at any rule setting. The boundary, reported with confidence intervals over hundreds of episodes, is the result. Where the boundary shows two hypotheses to be observationally equivalent, we treat the required extra channel as a physical variable to be determined by a distinguishability criterion, and we provide it.

1.3 Contributions

All contributions are methodological. **C1**: an operating-system-free C decision executive rendering the classical autonomy lineage as separable, heap-free layers. **C2**: an independent-V&V harness with a truth-holding physics referee, deterministic replay, and fault injection as

objective evidence. **C3**: a statistical observability-boundary method — distinguishability of hidden hypotheses under the sensor likelihood — exercised over a combined sweep with confidence intervals, showing three defenses share one indistinguishable region. **C4**: a cross-modal-consistency study that casts the full-signature decoy as an observational-equivalence region and reproduces the finding inside the ported C executive. **C5**: a minimal-observable formulation that asks not "which liveness sensor" but "which minimal additional observable set makes the target and decoy laws distinguishable," shows this set to be environment-dependent ($Z_{\min}(E)$), and delivers one strong candidate observable — a cardiopulmonary micro-Doppler phase — with governing physics and a measured separability, while framing the physically realizable airborne observable as an open sensing-design problem. A fidelity ladder and traceability keep every simulation number from being read as an operational claim.

2. Verification and Validation Framework

2.1 Software V&V and independent V&V

NASA software engineering requirements (NPR 7150.2) separate verification ("built right?") from validation ("the right thing for its environment?"), require both across the life cycle, and mandate bidirectional traceability [출처:조각]. The assurance and safety standard (NASA-STD-8739.8B) treats independent V&V as technical, managerial, and financial independence, and evaluates off-nominal behavior on objective evidence, not developer claims [출처:조각]. The Systems Engineering Handbook places V&V in the life cycle [출처:조각]. We keep verification and validation evidence in separate columns throughout.

2.2 Models and simulations as verifiable artifacts

NASA-STD-7009 requires models and simulations themselves to be verified, validated, and credibility-assessed before their outputs support decisions [출처:조각]. A simulation result is therefore verification evidence bounded by model fidelity — not operational validation. Flight practice reflects this: Mars 2020 terrain-relative navigation combined high-fidelity simulation with field and hardware-in-the-loop testing [출처:조각]; landing hazard-avoidance and precise-landing programs combined Monte-Carlo simulation, field tests, and HITL [출처:조각][출처:조각]; reference maps were rendered and then checked against real flight imagery [출처:조각]; the core Flight System exemplifies the component-plus-executive architecture we mirror [출처:조각].

2.3 The fidelity ladder

Every result is placed on the ladder of Table 1; this work sits at L2, and nothing here is presented as L3 or above.

Table 1. Fidelity ladder. L0–L2 are verification; L3–L5 are validation.

Level	Environment	Meaning	This work
L0	primitive	policy debug	host tests (synthetic obs.)
L1	procedural terrain	Monte Carlo	synthetic-DEM sweeps
L2	measured DEM + land-cover + physical sensor models	sensor simulation	← here (held-out Jirisan)
L3	DEM + real imagery + validated lighting	high-fidelity sim	not reached
L4	real terrain + real sensors	field validation	not reached
L5	real platform + sensors + software	system validation	not reached

3. The Reference World as Independent Referee

3.1 Three roles of one scene

Following NASA modeling-and-simulation practice, in which terrain, dynamics, sensors, actuators, and software are closed into one loop and onboard software is exercised against hidden truth [출처:조각], the reference world *sar/* is at once a **ground-truth generator** (hidden targets, decoys, and liveness on a measured DEM with land-cover), a **physical sensor simulator** (Section 3.2), and a **scenario generator**. The player never sees truth; it receives observations only, and a separate evaluator scores its actions against hidden state.

```

DEM (measured) + land-cover + scene materials
|   ground truth: hidden targets · decoys · liveness
physical sensor models (Planck · Koschmieder · LiDAR · GNSS · Audio · micro-Doppler)
|   synthetic observations   (player sees only this)
sensor-select → fusion → estimator (Kalman) → FDIR → planner → BT → arbiter → safety filter
|   action → independent evaluator scores vs hidden truth   ⌘

```

Fig. 1. Referee/player separation — the structural form of independent V&V.

3.2 Physical sensor models (Level-2 reference)

Each sensor is modeled at the level of its generative physics with representative literature parameters [출처:조각] (Appendix B). Optical photoelectrons with shot/read/PRNU noise and Koschmieder fog attenuation $\tau_{\text{vis}} = e^{-(3.912/V)R}$ (Eq. 1); LWIR Planck radiance over 8-14 μm with an atmospheric path and a smaller extinction than the visible band (Eqs. 2-3); LiDAR round-trip extinction with time-of-flight range (Eq. 4); a Gauss-Markov IMU (Eq. 5); GNSS position error from dilution and multipath (Eq. 6); and acoustic propagation through fog and canopy (Eq. 7).

$$N_e = \int E \cdot QE \cdot (\lambda/hc) d\lambda, \quad N_e \sim \text{Poisson}, \quad \tau_{\text{vis}} = \exp(-(3.912/V) \cdot R) \quad (1)$$

$$L_\lambda(T) = (2hc^2/\lambda^5) \cdot 1/(\exp(hc/\lambda k_B T) - 1) \quad (2)$$

$$L_{\text{sensor}} = \tau \cdot (\epsilon L_\lambda(T) + (1-\epsilon) L_\lambda(T_{\text{amb}})) + (1-\tau) L_{\text{path}}, \quad \beta_{\text{LWIR}} = (3.912/V)(0.15 + 0.65 e^{-V/150}) \quad (3)$$

$$P_r \propto \rho \cdot \exp(-2\beta R)/R^2, \quad R = c \cdot t/2, \quad \sigma_R \approx c \cdot t_{\text{pulse}}/(2\sqrt{\text{SNR}}) \quad (4)$$

$$a_m = (1+S_e)a + b + n, \quad \dot{b} = -b/\tau + w \quad (6) \quad \sigma_{\text{pos}} \approx \text{HDOP} \cdot \text{UERE} + \text{multipath} \quad (7) \quad \text{SPL}(r) = \text{SPL}_0 \quad (5) \\ 20 \log_{10} r - 0.005 r$$

The diversity is deliberate: one environmental change breaks each channel differently, so the policy selects by information quality rather than pinning a single sensor (Section 4.3).

4. Decision Executive Architecture

4.1 Superloop and blackboard

The executive is a single static blackboard and a heap-free superloop: sensors and health → selection → health-weighted fusion with temporal, cross-modal, and liveness gates → estimator → FDIR/supervisor → planner → behavior tree → arbiter → final safety filter. No dynamic allocation, no operating system; bounded memory and analyzable timing.

4.2 Estimation: constant-velocity Kalman filter and NIS

A per-axis two-state CV filter is kept in closed form ($F = [[1,1],[0,1]]$, $Q = q \cdot [[1/4, 1/2], [1/2, 1]]$):

$$\hat{x}^- = F \hat{x}, \quad P^- = F P F^T + Q \quad (9) \quad r = z - H \hat{x}^-, \quad S = H P^- H^T + R, \quad K = P^- H^T S^{-1}, \quad \hat{x} = \hat{x}^- + K r \quad (8)$$

$$d^2 = r^T S^{-1} r = r_x^2/(P_{x,00}+R) + r_y^2/(P_{y,00}+R), \quad d^2 > 50 \Rightarrow \text{re-init} \quad (10)$$

NIS is measured against the prediction as kinematic-consistency evidence; $\sigma = \sqrt{\text{tr } P}$ feeds the arbiter and the covariance guard. Code: Listing A.1.

4.3 Sensor selection policy (rule-based primary)

The **primary** selector is a rule that chooses, per condition, the sensor maximizing a detection-minus-cost-minus-energy objective:

$$U(a | s) = D(a, s) - \lambda \cdot C(a) - \mu \cdot E(a), \quad a^* = \underset{a}{\text{argmax}} U(a | s) \quad (11)$$

with $\lambda = 0.30$, $\mu = 0.10$, and $D(a, s)$ the referee's own condition-wise detection probability (so the label is proactive: condition $\rightarrow$ sensor $\rightarrow$ future detection, produced before observation). A NASA-S4 variant uses $U_i = \Delta P_i - \alpha \cdot \text{cost}_i$ with $\alpha = 0.35$ and a 0.05 hysteresis. A learned selective-state-space (Mamba) selector is reported as an ablation in Appendix C, not in the main path, so that the observability results do not depend on training choices.

4.4 Evidence fusion: soft vote, temporal, cross-modal, liveness

Fusion is health-weighted: the strongest present sensor anchors, present sensors within a gate are fused to a confidence-weighted centroid, and measurement noise decreases in confidence:

$$\text{anchor} = \underset{i:\text{present}}{\text{argmax}} w_i h_i c_i, \quad \text{conf} = \text{clamp}(\sum_{i \in \text{gate}} w_i h_i c_i, 0, 1) \quad (12)$$

$$(m_x, m_y) = (\sum w_i h_i c_i x_i, \sum w_i h_i c_i y_i) / \sum w_i h_i c_i, \quad R = 0.2 + 2.0(1 - \text{conf}) \quad (13)$$

$w = \{\text{RGB } 1.00, \text{ THERMAL } 0.70, \text{ LIDAR } 0.80, \text{ SAR } 0.50, \text{ AUDIO } 0.35\}$. Three gates follow: a kinematic gate drops associations with $d^2 > \gamma$; a cross-modal gate accepts by the count of distinct physical channels agreeing, with a conditional relaxation for obscuration-robust channels; and a liveness gate rejects candidates lacking a co-located vital-sign return:

$$\text{detected} = [n_{\text{modal}} \geq N_{\text{req}}], \quad N_{\text{req}} = 1 \text{ if } (N_{\text{cmpc}} \geq 2 \wedge \text{no blocked-channel} \wedge \text{some all-weather}) \\ \text{else } N_{\text{cmpc}} \quad (14)$$

$$\text{confirmed} = [\text{run}(\text{detected} \wedge \text{motion_ok}) \geq N] \wedge [\sigma \leq \sigma_{\text{max}}] \quad (15)$$

Code: Listing A.2; the shared cross-modal rule (Listing A.4) is used identically by prototype and experiments to prevent rule drift.

4.5 FDIR, arbitration, and the safety filter

A supervisor performs isolation/recovery over health, collision, power, actuator, and the covariance guard, with a priority FSM (Listing A.5). The arbiter admits precondition-satisfying actions and ranks them by $U(a) = \text{mission}(a) - 1.0 \cdot \text{collision_risk} - 0.3 \cdot \text{energy}(a) - 0.2 \cdot \sigma$, safety dominating utility (Eq. 16, Listing A.3). The final safety filter is unconditional (Listing A.6).

$$a^* = \underset{a : \text{precond}(a)}{\text{argmax}} U(a), \quad U(a) = \text{mission}(a) - 1.0 \cdot \text{collision_risk} - 0.3 \cdot \text{energy}(a) - 0.2 \cdot \sigma \quad (16)$$

4.6 Portability and the referee bridge

The decision core is portable C called from the referee through a foreign-function bridge that passes observations and liveness in and returns the chosen action out — so the referee scores the embedded code directly, and the C-port re-measurement of Section 7 is a measurement of the deployed rule.

5. Verification Methodology

Two techniques give requirements-based evidence. **Replay determinism**: a recorded stream replayed through a fresh executive reproduces the action sequence exactly. **Fault injection**: off-nominal conditions are injected and responses recorded; an uncaught fault is logged as identified risk, never coverage (Table 2).

Table 2. Injected faults and observed responses (off-nominal verification; fw/fault_test.c and the referee harness).

Injected fault	Observed response	Requirement	Status
Recorded scene replay	identical input yields identical action	R-FDIR-005	✓
Sensor dropout (health ↓)	isolation; fusion continues; recovery on restore	R-FDIR-001	✓
Covariance explosion	covariance finite; over-uncertain track not acted upon	R-FDIR-002	✓
Planner state corruption	recovers to a valid phase	R-FDIR-003	✓
Actuator unavailable	safe hold; event raised	R-FDIR-004	✓
Stale / stuck / biased sensor	not rejected — mistaken for a real target	GAP-FDIR-001	gap

The verification harness and tests are written and run separately from the implementation — an approximation of technical IV&V independence.

6. Observability: Definition and Boundary

6.1 A statistical definition (and what it is not)

We define observability as distinguishability of hidden hypotheses under the sensor likelihood. Let a hidden scenario s (e.g., a real target H_T or a decoy H_D) induce a law $P(Y_{0:T}|s)$ over the observation sequence. Two scenarios are **observationally equivalent** under an information set when

$$P(Y_{0:T} | s_1) = P(Y_{0:T} | s_2) \Rightarrow \text{no discriminator exceeds chance;} \quad (17)$$

if the laws differ on a set of positive measure, a discriminator may exist, its maximal power bounded by the total variation (equivalently the KL divergence) between the two laws. An observability boundary is then the partition of scenario pairs into distinguishable and observationally-equivalent under a given information set.

What this is not. This is not the control-theoretic observability of a linear system, whose observability matrix $O = [H; HF; HF^2; \dots; HF^{n-1}]^T$ tests whether the state x can be reconstructed from the outputs of a known model. Here we test whether two hypotheses can be separated under a nonlinear, stochastic sensor likelihood — a detection/hypothesis-testing notion, not state reconstruction. We use "observability" only in this statistical sense.

Operationalization (honest). We cannot evaluate $P(Y|s)$ in closed form, so we estimate distinguishability by the performance a specific discriminator (the fusion rule) achieves over Monte-Carlo episodes: false-alarm and detection rates, with confidence intervals, are empirical lower bounds on distinguishability. If even the best rule we try leaves the false-alarm rate at the clutter level, that is evidence — not proof — of near-equivalence $P(Y|s_1) \approx P(Y|s_2)$; if adding a channel collapses it, that is evidence the joint law becomes distinguishable. The information sets I_0, I_1, I_2 of Section 7 are thus a sequence in which the indistinguishable region shrinks.

6.2 Three orthogonal defense axes

Table 3. The three defense axes: what each rejects, what each cannot, and its cost.

Axis	Mechanism	Rejects	Cannot reject	Cost
Temporal (window N)	same location for N steps	transient clutter	clutter persisting $\geq N$ steps	detection ↓, latency ↑
Kinematic (NIS)	innovation d^2 vs track	jumping / impossible motion	stationary false target	false negatives on sharp turns
FDIR	health, collision, power, actuator, covariance	dropout, power, collision, actuator, blow-up	stale / stuck / biased sensor	—

6.3 Combined sweep with confidence intervals

A single sweep over confirmation window (N), clutter persistence, and target motion places all three boundaries on one table. On the held-out Jirisan measured DEM with clutter density 0.20, each cell reports detection rate and mean false-alarms per episode with 95% confidence intervals over $N = 100$ episodes (`sar/ivv/fw_vv.py --mode grid`).

Table 4. Combined sweep, $N = 100$ episodes/cell. Top line: detection % (95% CI). Bottom line: false alarms/episode (95% CI). Clutter = 0.20.

motion · persistence	N = 1	N = 2	N = 3
static · transient	4% (1-8) 2.47 (2.25-2.69)	16% (10-22) 0.65 (0.50-0.80)	4% (1-8) 0.16 (0.07-0.25)
static · persistent	6% (2-11) 3.07 (2.83-3.31)	10% (6-15) 2.84 (2.61-3.07)	10% (5-15) 2.59 (2.36-2.82)
const-vel · transient	12% (6-17) 2.55 (2.33-2.77)	19% (12-26) 0.71 (0.55-0.87)	8% (4-13) 0.18 (0.09-0.27)
const-vel · persistent	8% (4-13) 3.02 (2.79-3.25)	8% (4-12) 2.85 (2.62-3.08)	12% (6-17) 2.44 (2.21-2.67)

The confidence intervals make the boundaries, not the individual cells, the claim. Across N: from $N = 1$ to $N = 2$ the transient false-alarm rate falls with non-overlapping intervals (static 2.47 [2.25–2.69] → 0.65 [0.50–0.80]; const-vel 2.55 → 0.71), while detection stays low. Across persistence: once clutter persists, the false-alarm rate stays near 2.4–3.1 at every N, intervals well above the transient $N = 2$ rate — the temporal defense is gone. Across motion: detection intervals are wide and overlapping, so we claim only the direction (moving targets no easier), not per-cell values. The persistent-clutter columns are the boundary: no confirmation window closes them.

6.4 The shared indistinguishable region

The three axes are complementary in what they close and identical in what they cannot. A stationary, kinematically consistent, persistent false signal passes all three, because in time,

motion, and health it induces the same observations as a real target (Fig. 2) — an observationally-equivalent region in the sense of Eq. (17).

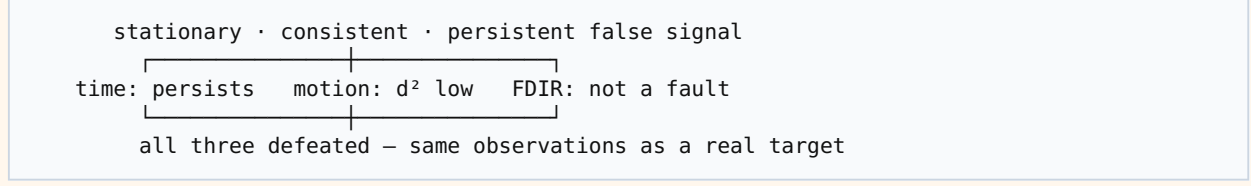


Fig. 2. The shared indistinguishable region; a different channel is required.

7. Cross-Modal Consistency and Observational Equivalence

7.1 Method: widen the information set, hold the sensor count

We change fusion from spatial voting to cross-modal consistency, keeping five sensors. Detection is decoupled from navigation: one fixed-path observation stream is recorded per scenario and replayed under each information set, applying only the claim rule (true replay). We evaluate on held-out Jirisan with $N = 200$ episodes (sar/ivv/cmpc_vv.py). Information sets: **I0** spatial ($\geq$ one modality); **I1s** strict CMPC ($\geq$ two distinct channels); **I1c** CMPC with the conditional relaxation of Eq. (14); **I2** CMPC-cond plus a required liveness channel. Scenarios: A single-modality clutter, B partly-observed real target, C full-signature decoy.

7.2 Results with confidence intervals

Table 5. Prototype rule (Python), $N = 200$ episodes. FA rows: mean false alarms/episode (95% CI) [episodes with ≥ 1 FA]. Row B: detection % (Wilson 95% CI).

Scenario	I0 spatial	I1s strict	I1c cond	I2 + liveness
A clutter (FA/ep)	2.92 (2.76–3.07) [200/200]	0.94 (0.85–1.03) [151/200]	2.02 (1.90–2.15) [199/200]	0.03 (0.01–0.05) [6/200]
B real (det %)	95% (91–97)	16% (12–22)	52% (46–59)	47% (40–54)
C decoy (FA/ep)	1.71 (1.54–1.88) [200/200]	1.02 (0.98–1.06) [195/200]	1.71 (1.54–1.88) [200/200]	0.07 (0.03–0.12) [13/200]

Table 6. Deployed C executive (fw/evidence.c) on identical streams, $N = 200$. C implements the conditional CMPC only.

Scenario	I0 spatial	I1c cond	I2 + liveness
A clutter (FA/ep)	1.76 (1.68–1.85) [200/200]	1.07 (0.99–1.15) [172/200]	0.00 [0/200]
B real (det %)	95% (91–97)	53% (46–60)	46% (39–53)
C decoy (FA/ep)	0.92 (0.88–0.95) [183/200]	0.92 (0.88–0.95) [183/200]	0.00 [0/200]

7.3 Reading the boundary

- Single-modality clutter (A) is distinguishable with five sensors.** Strict CMPC drops false alarms $2.92 \rightarrow 0.94$ (intervals do not overlap) — no new sensor.
- At a measured cost (B).** A real target seen only through SAR under canopy and fog falls $95\% \rightarrow 16\%$ under strict CMPC; the conditional relaxation recovers it to **52% (46–59)** — not the 60% a small early sample suggested, a correction the larger N makes visible.
- The relaxation reopens a clutter hole (A: 2.02 [1.90–2.15]).** Suppressing single-modality clutter and preserving single-modality real targets cannot be done together with modal rules alone — a trade.
- The full-signature decoy (C) is observationally equivalent under five channels.** CMPC does not move it (I0 1.71 vs I1c 1.71 in the prototype; 0.92 vs 0.92 in C — overlapping intervals). Formally, there is a region where

$$I = \{y_{0:T}^{\text{RGB}}, y_{0:T}^{\text{LWIR}}, y_{0:T}^{\text{LiDAR}}, y_{0:T}^{\text{SAR}}, y_{0:T}^{\text{Audio}}\}, \quad P(I | H_T) \approx P(I | H_D) \quad (18)$$

so no rule on I separates target from decoy. Adding a liveness channel z makes the joint law distinguishable, and the false-alarm rate collapses:

$P(I, z | H_T) \neq P(I, z | H_D) \Rightarrow$ C decoy: $1.71 \rightarrow 0.07$ [13/200] (prototype), $0.92 \rightarrow 0.00$ [0/200] (C)

On the zeros. The C executive gives 0/200 episodes with a decoy false alarm; by the rule of three the 95% upper bound is $\approx 3/200 \approx 1.5\%$ of episodes, not "perfect." The prototype does not reach zero — it leaves 0.07 FA/ep [13/200], because it scores every cluster statelessly while the single-hypothesis C tracker claims distant clutter less often. Rule equivalence (host test [14]), directional reproduction, and this structural magnitude difference are three separate statements, kept apart.

8. The Minimal-Observable Problem: What Must Be Added, and Under Which Environment

8.1 The question, reframed

Section 7 shows the required channel is a discriminating one, not a redundant locator. Following NASA autonomy-V&V practice, which treats sensing, uncertainty, and action as one system rather than seeking a single perfect sensor [출처:조각], we do not ask "which liveness sensor to add." We ask: **what is the minimal additional observable set that makes the target and decoy laws statistically distinguishable, and does that set depend on the environment?** The purpose of the experiment is not to propose a specific liveness sensor, but to determine whether an additional physical observable is required for discrimination — and, if so, which.

8.2 Motion is not liveness; thermal is not liveness

At a hundred metres, a camera sees motion that may be a walking person or wind-driven vegetation; a thermal channel sees a contrast that may be a person or a sun-warmed rock, a vehicle, a roof, or an animal. Hence motion $\neq$ liveness and thermal $\neq$ liveness; stating this as a limitation is a strength. The observed velocity is a sum, so a fixed motion threshold fails once wind lifts the vegetation term:

$$v_{\text{obs}} = v_{\text{target}} + v_{\text{platform}} + v_{\text{wind}} + \varepsilon \Rightarrow \text{the discriminating power of any } z \text{ is} \quad (20)$$

conditioned on the environment: $P(z | H, E)$, not $P(z | H)$

with $E = \{\text{wind, illumination, altitude, view angle, vegetation density, terrain}\}$. NASA UAS detect-and-avoid V&V likewise models sensor uncertainty and sweeps scenario parameters rather than assuming an ideal sensor [출처:조각].

8.3 An observable stack and incremental distinguishability

We build observables in layers — **L0** single-frame (appearance, thermal contrast, shape); **L1** temporal (displacement, velocity, motion persistence, motion periodicity, articulated motion, motion-background independence, multi-view consistency) — and add them one at a time, measuring how much each raises the separation of the two laws. With D a divergence (KL / Jensen-Shannon) or a likelihood-ratio separation, the minimal sufficient set is the smallest layer at which D first exceeds the required separation, and it is a function of the environment:

$$D_k(E) = D(P(I_0, z_{1:k} | H_T, E) \| P(I_0, z_{1:k} | H_D, E)), \quad Z_{\min}(E) = \underset{D_{\text{req}}}{\operatorname{argmin}}_Z |Z| \text{ s.t. } D_{|Z|}(E) \geq D_{\text{req}}$$

The full experimental design is therefore a product, not a single "human vs. tree" contrast:

Scenario = Target × Environment × Sensor × Geometry	
Target	: stationary / walking / partly-occluded human
Decoy	: vegetation · thermal rock · animal · moving object · static heat source
Environment	: no / weak / strong wind · sunlight · shadow · thermal-background variation
Geometry	: altitude · view angle · target size · occlusion · range

Fig. 3. The observability experiment as a product design; $P(I|H_T, E)$ and $P(I|H_D, E)$ are measured per cell. This report populates a slice (Sections 6–7 and 8.5); the full grid is the defined program.

8.4 Candidate observables and the confounders

No single-frame or purely-kinematic observable is liveness: decoys share motion, thermal contrast, and persistence with a real target (Table 8), which is exactly why the three defenses of Section 6 leave a shared region. Structure begins to separate the laws only in the temporal and cross-sensor layers, and the strongest single candidate we identify — a non-contact, obscurant-penetrating, localizable observable — is an RF micro-Doppler cardiopulmonary phase (FINDER-class [출처:조각]).

Table 7. Candidate observables against the airborne constraints (non-contact, penetrating obscurants, localizable).

observable	non-contact	penetrates	localizable	note
appearance / shape (L0)	yes	no	yes	collapses under occlusion, night, small footprint
thermal contrast (L0)	yes	partial	yes	shared by rock/vehicle/animal — not liveness
motion structure (L1)	yes	partial	yes	periodicity/articulation/ background-independence; confounded by wind
exhaled CO ₂	yes	weak	none	diffuses; no bearing → cannot localize
contact/seismic vibration	no	n/a	medium	needs coupling → unfit for an airborne UAV
RF micro-Doppler (cardiopulmonary)	yes	strong	strong	★ non-contact, penetrates rubble/ canopy, localizable

Table 8. Confounder matrix — which observable each source excites. ✓ present · ~ conditional · ? structure-dependent · × absent. Motion, thermal, and persistence are shared, so none alone is liveness.

observable	human	vegetation	thermal rock	animal	static heat
motion	✓	✓ (wind)	×	✓	×
thermal contrast	✓	~	✓	✓	✓
persistence	✓	✓	✓	✓	✓
motion periodicity	?	? (wind)	×	?	×
articulated motion	?	×/?	×	?	×
micro-Doppler (vital)	✓	×	×	✓	×

Liveness is not identity. The micro-Doppler vital-sign observable separates live from non-live, but an animal is also live (last column) — so it distinguishes a person from vegetation, rock, and heat sources, **not** a person from an animal. Human identification needs a further observable (e.g., gait or articulation structure), which we do not claim to resolve.

8.5 A measured instance: imaging cannot separate the decoy; adding micro-Doppler can

The imaging information set already gives one point of the incremental curve: for the full-signature decoy the best rule on I does not move the false-alarm rate (Tables 5–6, row C: $I_0 \approx I_1$), so $D_0 \approx 0$ — near-equivalent laws. Adding the micro-Doppler observable raises the separation sharply. Its physics — chest-wall micro-motion read as a radar phase, with the band energy of the unwrapped phase in the cardiopulmonary band as the statistic:

$$x(t) = A_r \sin(2\pi f_r t) + A_h \sin(2\pi f_h t) \quad (23) \quad \varphi(t) = (4\pi/\lambda) x(t) \quad (22)$$

$$s(t) = e^{j\varphi(t)} + n(t), n \sim \text{CN}(0, \sigma^2) \quad (25) \quad z = \int_{f_{lo}}^{f_{hi}} |\mathcal{F}\{\angle s\}|^2 df / \int |\mathcal{F}\{\angle s\}|^2 df \quad (24)$$

with literature parameters [출처:조각] $f_r \approx 0.2\text{--}0.34$ Hz, $f_h \approx 1.0\text{--}1.7$ Hz, band $[0.15, 2]$ Hz. A non-live decoy excites the five imaging channels but has no cardiopulmonary periodicity, so z stays at the noise floor. Modeling z in `sar/liveness_ref.py` over $N = 400$ episodes gives, at a representative post-integration SNR of 6 dB, **AUC = 0.977** and symmetric **$D_{KL} \approx 16.5$ nats** (band statistic 0.583 ± 0.175 under H_T vs 0.193 ± 0.054 under H_D); two independent estimators agree (Table 9).

Table 9. Separability of the micro-Doppler observable vs. SNR ($N = 400/\text{point}$) — one environment axis. The separation is the signal, not an easy operating point: near chance at low SNR.

post-integration SNR	0 dB	3 dB	6 dB	12 dB
ROC-AUC	0.53	0.69	0.98	1.00
D_{KL} (sym, nats)	0.00	0.28	16.5	147

Honest operating point. At 0 dB the AUC is 0.53 — chance — so the high separability is not an artifact of an easy setting; it emerges only once the phase SNR reaches a few dB. This SNR sweep is one axis of E ; the full $Z_{\min}(E)$ over wind, illumination, altitude, view angle, and vegetation is not yet measured.

8.6 Result and open problem

The result is not a liveness sensor but a boundary and a program. An additional observable is required; a cardiopulmonary micro-Doppler phase is a strong candidate in the modeled regime; but **there may be no universally sufficient liveness parameter** — the minimal sufficient set is environment-dependent, $z^*(E) = f(E)$, a set $Z_{\min}(E)$. Identifying the physically realizable observable that provides sufficient discrimination from an airborne platform, across wind, illumination, altitude, view angle, and vegetation, remains an open sensing-design problem. The liveness channel here is a modeled physical variable, not a field-validated sensor, and it resolves live-versus-non-live, not human-versus-animal.

9. Traceability

Table 10. Cross-modal, FDIR, and liveness traceability. ✓ = obtained (verification, L2); ○ = remaining (field validation).

ID	Requirement / item	Status	Evidence
R-CMPC-001	persistent observation not accepted on a single channel	✓	Table 5; test_cmpc
R-CMPC-002	obscuration-robust-only real target not discarded (conditional)	✓	Table 5 row B
R-CMPC-003	full-signature false target not accepted without liveness	✓	Tables 5–6 row C
R-LIVE-001	a candidate observable separates live H_T from non-live H_D (modeled regime)	✓	Table 9; sar/liveness_ref.py
R-LIVE-002	minimal sufficient observable set $Z_{\min}(E)$ across environments	○	§8.3, §8.6 (open program)
FW-CMPC-001	flight/embedded port (C), rule preserved	✓	fw/evidence.c; host_test.c [14]
R-FDIR-001...005	isolation / guard / recovery / safe-hold / determinism	✓	Table 2; fw/fault_test.c
GAP-FDIR-001	stale / stuck / biased sensor not rejected	risk	Table 2; §6.4
V-FIELD-*	target / high-fidelity / field validation; z_s realizability & minimality	○	— (companion plan; §8.5)

10. Limitations and Negative Results

- **Simulation only.** Every number is L2 verification; sensor, decoy, and liveness models are representative physics, not validated against real sensors.
- **The gap is real.** The stationary-consistent-persistent region is unresolved by time, motion, or fault management; liveness is required and is modeled, not field-validated.
- **The false-alarm floor was once an artifact.** A clutter-free early configuration reported zero false alarms — a property of the scene, not the policy; adding clutter exposed the temporal limit of Table 4.
- **The conditional relaxation is a trade, not a fix,** and the larger sample corrected its recovered detection from an early 60% to 52% (46–59).
- **Zeros are bounded, not perfect.** A 0/200 cell has a rule-of-three 95% upper bound near 1.5% of episodes; the prototype liveness column is not zero (0.07 [13/200]).
- **The learned selector is an ablation,** moved to Appendix C so the observability results do not rest on training choices; its earlier feedback feature caused an RGB-lock-in covariate shift, corrected by a condition-only, proactive label.
- **Single hypothesis.** Detection is capped and multi-target scenes under-served — a measured design choice that also explains the lower C false-alarm magnitudes.
- **Motion and thermal are not liveness.** Wind-driven vegetation and sun-warmed terrain confound them; the required observable is environment-conditioned, $P(z|H,E)$, and no single universal liveness parameter is claimed.
- **Only a slice of the design was run.** The Target $\times$ Environment $\times$ Sensor $\times$ Geometry grid (Fig. 3) is measured only in part; the environment-dependent minimal set $Z_{\min}(E)$ and the physical realizability of any airborne liveness observable are open (§8.6), and the micro-Doppler channel resolves live-vs-non-live, not human-vs-animal.

11. Concluding Remarks

Giving "observability" a statistical meaning turns an engineering story into a testable one: temporal persistence, kinematic consistency, and fault management each separate a distinct false-signal class, but leave one observationally-equivalent region; cross-modal consistency separates single-channel clutter without new hardware, at a measured cost; and a full-

signature decoy stays observationally equivalent under the five imaging channels until a discriminating observable is added. We do not stop at "add a liveness sensor": we pose the minimal-observable problem, show that motion and thermal are not liveness and that the sufficient observable set is environment-dependent ($Z_{\min}(E)$), and offer one strong candidate — a cardiopulmonary micro-Doppler phase — with governing physics and a measured separability that honestly collapses at low SNR and that resolves live-vs-non-live rather than identity. Which physically realizable airborne observable suffices under each environment is left as an explicit open sensing-design problem. Every result is simulation verification at a stated fidelity level, with confidence intervals and with validation kept separate and explicitly not yet done.

Appendix A — Deployed Code Listings

Excerpts from the deployed C decision core and the shared cross-modal rule; comments abbreviated, logic verbatim.

Listing A.1 — CV Kalman update and NIS (fw/estimator.c)

```
float fw_kf_update(KalmanCV *kf, float zx, float zy, float r){
    if(!kf->init){ axis_init(...); kf->init=true; return 0.0f; }
    float rx=zx-kf->xp, ry=zy-kf->yp; /* NIS vs prediction */
    float d2 = rx*rx/(kf->Px[0][0]+r) + ry*ry/(kf->Py[0][0]+r); kf->nis=d2;
    if(d2 > KF_REINIT_D2){ axis_init(...); return d2; } /* lost/new target */
    axis_update(&kf->xp,&kf->xv,kf->Px,zx,r); axis_update(&kf->yp,&kf->yv,kf->Py,zy,r);
    return d2; }
```

Listing A.2 — Health-weighted fusion: CMPC, liveness, temporal, guard (fw/evidence.c)

```
float conf = clampf(num,0,1); bool detected = (conf >= th); /* CMPC off: soft vote */
float d2 = fw_kf_update(&bb->kf, xw/wsum, yw/wsum, 0.2f+2.0f*(1.0f-conf));
if(bb->cfg.nis_gate>0 && d2>bb->cfg.nis_gate) motion_ok=false; /* kinematic gate */
if(bb->cfg.cmpc_min>0){ int n_modal=0; for(i=0;i<FW_N_SENSOR;i++) n_modal+=in_gate[i];
    int required=bb->cfg.cmpc_min;
    if(cmpc_min>=2 && !any_blocked && any_allw) required=1; /* conditional relax */
    detected = (n_modal >= required); }
if(bb->cfg.require_live && !(bb->live.present && dist2(live,m)<=FW_GATE2)) detected=false;
if(detected && motion_ok) bb->belief.confirm_run++; else bb->belief.confirm_run=0;
bb->belief.confirmed = (bb->belief.confirm_run>=need) && (bb->belief.sigma<=FW_SIGMA_MAX);
```

Listing A.3 — Arbiter utility (fw/arbiter.c) · A.4 — cross-modal rule (policies.py)

```
float fw_utility(bb,a){ return mission(a) - 1.0f*collision - 0.3f*energy(a) - 0.2f*sigma; }
def cmpc_confirm(obs,cmpc_min=2,require_live=False,gate=2.5,conditional=True):
    for cl in clusters:
        mods=set(per); required=cmpc_min
        if conditional and cmpc_min>=2 and mods.isdisjoint(BLOCKED) and mods&ALLWEATHER: required=1
        if len(mods)<required: continue
        if require_live and not any(near(live,cl)): continue # liveness gate
        out.append((wx,wy,min(1.0,len(mods)/3.0)))
```

Listing A.5 — FDIR FSM (fw/safety.c) · A.6 — safety filter

```
if(collision>=HARD||batt<EMERG) safety=EMERGENCY; else if(batt<RETURN) safety=RETURN;
else if(!imu_valid||!gps_valid||dead>=2) safety=DEGRADED; else safety=NORMAL;
Action fw_safety_filter(bb,proposed){ if(!actuator_ok) return NONE; if(collision>=HARD) return AVOID;
    if(batt<EMERG) return EMERGENCY; if(batt<RETURN) return RETURN; if(!imu_valid) return RELOCALIZE;
    if((proposed==APPROACH||proposed==INSPECT)&&collision>=0.5f) return SEARCH; return proposed; }
```

Listing A.7 — Liveness variable z_5 (sar/liveness_ref.py)

```
def microdoppler_baseband(t, x, fc_hz, snr_db, rng):
    lam = C_LIGHT/fc_hz; phi = (4*pi/lam)*x          #  $\varphi=(4\pi/\lambda)x$ 
    return exp(1j*phi) + CN(0, 10**(-snr_db/10))
def band_energy_stat(s, fs, f_lo=0.15, f_hi=2.0):
    phi = unwrap(angle(s)); P = |rfft(phi-mean)|**2
    return P[f_lo:f_hi].sum() / P.sum()                #  $z$  = cardiopulmonary band ratio
```

Appendix B — Sensor Model Parameters

Table B.1. Representative parameters of the Level-2 sensor reference models [출처:조각]. Not field-validated.

Sensor	Reference class	Governing relation	Key parameters
RGB	ISETCam	$N_e = fE \cdot QE \cdot \lambda / hc$; $\tau_{vis} = e^{-(3.912/V)R}$	full-well 10000 e ⁻ , read 2 e ⁻ , PRNU 0.01
Thermal LWIR	DIRSIG+MODTRAN	Planck Eq.(2); L_{sensor} Eq.(3)	8-14 μ m, NETD 0.05 K, ε 0.95
LiDAR	DIRSIG waveform	$P_r \propto \rho e^{-2\beta R/R^2}$; $R = ct/2$	pulse 5 ns, tx 10 ⁹ photons, det-eff 0.4
IMU	IEEE Allan variance	$a_m = (1+S)a + b + n$; $\dot{b} = -b/\tau + w$	ARW 0.02, bias-instab 0.01, τ 100 s
GNSS	ESA receiver	$\sigma \approx HDOP \cdot UERE + \text{multipath}$	UERE 3 m; multipath 8/20 m urban/canyon
Audio	image-source RIR	$SPL(r) = SPL_0 - 20\log_{10}r - 0.005r$	c 343 m/s, 4 mics, 0.3 m spacing
Liveness (z_5)	micro-Doppler (FINDER-class)	$\varphi = (4\pi/\lambda)x$; Eqs.(21-24)	f_r 0.2-0.34 Hz, f_h 1.0-1.7 Hz, f_c 3 GHz

Appendix C — Learned Sensor-Selector Ablation

A selective-state-space (Mamba) selector reads the observation context ([visibility, IMU-ok, GNSS-uncertain]) and chooses the sensor to trust. It is an ablation, not the reported system: the observability results of Sections 6-8 use the rule-based selector of Eq. (11) so they do not depend on training set, hyperparameters, model size, or leakage. The one reportable negative result: an earlier feature that fed the previous choice back induced a self-reinforcing covariate shift (an RGB→RGB lock-in); it was removed and the training label made proactive and condition-based (condition → sensor → future detection). A full learned-vs-rule comparison — generalization, inference cost, and baselines — is deferred; nothing in Sections 6-8 depends on it.

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